# LÉHO DEHOVE

lehodehove@gmail.com | +33 06 15 50 16 95 | linkedin.com/in/leho-dehove | dehoveleho.github.io

## Engineering Experience

**CNRS – ESPCI PSL University (Brain Plasticity Lab)** June – expected Sept. 2026
**Computer Vision R&D Intern**

- Developed **KiAnSo**, an automated video processing software in Python for Drosophila behavioral neuroscience, replacing manual tracking, increasing analysis throughput by **+500%**, and saving **over 45 hours per batch**.
- Engineered **OptoStim**, an automated optogenetic photostimulation hardware bench and high-speed synchronized video acquisition platform.
- Co-authoring a research publication on automated neural locomotion kinematics under optical stimuli.

**Danfoss Power Solutions (Manufacturing Plant)** June – July 2025
**Manufacturing & Quality Operator**

- Precision metrology, quality inspection, and assembly of mechanical and hydraulic components for fluid power systems.

**ESME FabLab & Robotics Club** Sept. 2022 – June 2025
**FabLab Administrator & Club Vice-President**

- Managed digital fabrication machines (CNC milling, SLA/FDM 3D printing, laser cutting) and trained **150+ engineering students** in CAD/CAM and rapid physical prototyping.

**Universiti Putra Malaysia (UPM)** 2025 – 2026
**Academic Exchange – Electrical & Electronic Engineering**

- International coursework and projects in control systems, sensor fusion, and embedded architectures. English C1.

## Projects: Computer Vision & Hardware Systems

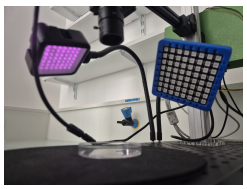


### KiAnSo – Kinematic Analysis Software

CNRS / ESPCI | Sept. 2026

*Python, OpenCV, Farnebäck Optical Flow, Savitzky-Golay, Hampel Filtering, NumPy, SciPy*

- Automated kinematic analysis pipeline built to replace manual, subjective frame-by-frame scoring in Drosophila behavioral neuroscience.
- Implemented Farnebäck dense optical flow, robust signal filtering (Hampel outlier rejection and polynomial smoothing), and automated ROI segmentation to model embryonic spasms and adult locomotion.
- Achieved a **+500% throughput increase**, turning a 5-day week-long task into a single day and saving **over 45 hours per experimental run**.







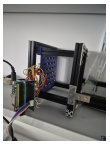
### OptoStim – Optogenetics Bench & Video Acquisition

CNRS / ESPCI | Sept. 2026

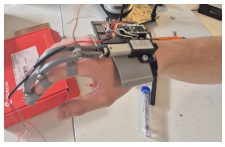
*Python, Flask, Custom LED Driver Matrix, Extruded Aluminum, Microcontrollers*

- Automated photostimulation and high-speed video recording bench for behavioral neural tracking under optical stimulation.
- Engineered modular extruded aluminum frame, custom LED driver board, and microcontroller firmware delivering programmable optical pulse trains.
- Developed Python/Flask acquisition server coordinating camera exposure triggers with excitation pulses, directly coupled to KiAnSo for automated kinematic extraction.





## Bio-Mechatronics, Exoskeletons & Physical Robotics

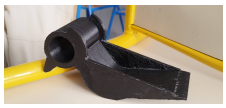



### AILAPS – Active Hand Exoskeleton for Rehabilitation

May 2026

*Biomechanics, SolidWorks, Embedded C, Actuation, Additive Manufacturing*

- Motorized wearable hand orthosis prototype assisting patients with everyday muscular weakness and finger flexion/extension rehabilitation.
- Designed compliant multi-bar linkages in SolidWorks to follow anatomical finger trajectories without joint hyperextension.
- Programmed deterministic embedded control in C, integrated power drivers, and fabricated high-strength 3D printed mechanical enclosures.

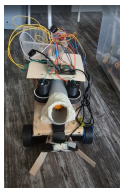



### Cuicui – Pediatric Upper-Limb Mechanical Prosthesis

June 2023

*Biomechanics, Ergonomics, Alginate/Plaster Casting, Passive Mechanics, 3D Printing*

- Custom mechanical prosthetic device developed for a young child with congenital upper limb agenesis to ride a bicycle independently.
- Cast direct anatomical limb molds using medical alginate and dental plaster to guarantee a comfortable socket without painful pressure points.
- Engineered a passive, fail-safe handlebar clamping and quick-release mechanism providing stable bicycle steering with zero electronic dependency.




### Scotchouille – Autonomous Mobile Rover (Tournament Winner)

June 2025

*Python, OpenCV, Differential Drive, Trajectory Planning, Flywheel Ballistics*

- Autonomous competition robot designed to detect, track, retrieve, and shoot targets on an arena under strict match rules. **Won 1st place in tournament**.
- Implemented monocular OpenCV computer vision for live ball detection and kinematic trajectory calculation on a custom differential drive chassis.
- Built high-velocity flywheel projectile launch mechanism with servo-driven feed and dedicated battery distribution.



### Louya – Custom Direct-Drive 3D Printing Testbed

2016 – Present

*Additive Manufacturing, Direct-Drive, Hotend Design, Firmware Tuning, DFM/DFA*

- Modified Creality CR-10 testbed upgraded over 8+ years: custom direct-drive extrusion, all-metal hotend, mechanical alignments, and tuned Klipper/Marlin firmware.
- Continuous testbed for technical polymers (PLA, PETG, TPU, Nylon, Carbon-Fiber composites) and Design for Additive Manufacturing (DFAM).

## Control Theory & State Estimation Experiments

### Hybrid Neural vs PID Control (Quadrotor Payload)

*Control Theory, 2-layer MLP in C, ARM Cortex-M4, NMPC*

- Addressed underactuated nonlinear swing in suspended quadrotor payloads during aggressive maneuvers ($> 20^\circ$ tilt) where classical linear PID gains saturate.
- Trained a compact 2-layer MLP (106 parameters, 8 linear algebra equations in C, **evaluates in 2.5 $\mu$s** on Cortex-M4).
- **Result:** Hybrid architecture (PID integral baseline + neural policy) **cut tracking error by 43%** while maintaining zero steady-state wind drift.

### 3D State Estimation with Orocos BFL

*C++, Orocos BFL, Extended Kalman Filter, Particle Filter, OpenCV*

- Benchmark of Bayesian estimators on rapid deterministic 3D trajectories (4D state: $[z, \dot{z}, \theta, \omega]$ at 6 Hz spin) using OpenCV and Orocos BFL.
- **EKF:** Executed in **0.04 ms/step**, converged onto 37.7 rad/s spin in 5 frames (**96.4% prediction accuracy** in 200 ms).
- **Particle Filter (SMC):** 4.2 ms/step, handled multi-modal distributions through visual occlusions and verified filter confidence drops under ill-conditioned camera angles.